Noirit Ghosh Choudhuri

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EDUCATION

University of California, Berkeley · Berkeley, CA
B.A., Computer Science · Expected Graduation: 2029
Current coursework: Structure and Interpretation of Computer Programs (CS 61A), Linear Algebra and Differential Equations (MATH 54)

The Ohio State University · Columbus, OH
GPA: 4.0 · 2025-2026
Relevant coursework: Multivariable Calculus, Linear Algebra, and Software Components

EXPERIENCE

Lead Developer
Beat-Less Tuning App *January 2025 - Present, Columbus, OH*
- Developed an iOS just-intonation training app using React Native, Expo, and JavaScript, designed to help professional musicians and students develop accurate pitch perception and intonation.
- Engineered a custom audio playback system using expo-audio with real-time pitch/rate control, looping, and pitch-shifted drones, and built interactive tuning features including a circular cent-based interface, chord generation, and multiple ear-training practice modes.
- Published the app to the App Store, generating **400+ copies sold and $2,000+ in revenue.**

Internal Vice President & Software Lead
VINdicated *August 2026 - Present, Berkeley, CA*
- Led development of a used-car pricing education game and contributed to its Python/FastAPI backend, supporting **800 playable rounds**, price-guess scoring, daily challenges, and persistent player statistics.
- Led development of a CatBoost valuation pipeline using **21,500+ vehicle records**, achieving **4.5% median absolute percentage error** against CARFAX estimates on **4,306 held-out vehicles**.
- Guided development of SHAP-based price explanations and mileage and accident adjustments; managed recruitment, finances, and chapter logistics.

Lab Assistant
SOAR Lab @ Ohio State University *January 2026 - May 2026, Columbus, OH*
- Programmed and assembled a swarm of 6 Crazyflie nano-drones using Python and ROS, achieving synchronized autonomous flight paths in a lab research setting.
- Integrated a Vicon motion-capture system for real-time position tracking, enabling precise closed-loop control across all 6 drones simultaneously.
- Implemented hardware kill-switch safety protocols to ensure safe operation during synchronized multi-drone flight testing.

PROJECTS

Buzzlytix
Full-Stack Development • buzzlytix.com *May 2025 - Present*
- Engineered a concurrent Python scraping and ETL pipeline spanning **165 RSS feeds across 71 configured news sources**, implementing data extraction, cleaning, deduplication, and persistent storage in SQLite.
- Deployed cloud data infrastructure using **Google Cloud Run and Google Cloud Storage**, automating dataset updates and integrating processed news data with the production web application.
- Built an AI-powered data processing engine that clusters news into **up to 15 ranked topics per run**, generating source-cited summaries and cross-publisher framing comparisons with structured-output validation, caching, and automated retries.
- Developed a responsive **Next.js, React, and TypeScript** platform featuring interactive topic navigation, publisher coverage analytics, and side-by-side source comparisons.
- Tech: Python, TypeScript, JavaScript, React, Next.js, OpenAI API, SQLite, RSS, Web Scraping, ETL, Data Processing, Sentiment Analysis, Topic Clustering, GCS, Cloud Run, Docker, REST APIs.

Geo-Feature Sharing Platform Demo
Full-Stack Development • github.com/noiritgc/geo-feature-sharing-platform *January 2026 - Present*
- Engineered a full-stack geospatial web application using Next.js, React, TypeScript, Leaflet, and Firebase, integrating interactive map features with Google Authentication and Firestore to model users, friends, groups, and saved geographic areas through relational references.
- Implemented interactive polygon drawing and geolocation, enabling users to define and save geographic regions through a Leaflet interface.
- Tech: Next.js 16, React 19, TypeScript, Firebase (Authentication, Firestore, Analytics), Leaflet, React-Leaflet, CSS Modules, ESLint.

Public Audio Synthesis Toolset
Python Programming & Signal Processing • github.com/noiritgc/Audio-Processing-Generation *February 2025 - Present*
- Developed a Python audio synthesis pipeline using NumPy, Librosa, and WORLD (pyworld), applying Fourier analysis to extract spectral envelopes and timbral characteristics from real instrument recordings.
- Implemented additive synthesis to reconstruct **60 instrument tones from octaves 2-6** at mathematically precise pitches while preserving instrument resonance and eliminating pitch-shifting artifacts.
- Tech: Python, NumPy, Librosa, PyWorld, SoundFile, FFmpeg.

SKILLS

Languages: Python, TypeScript, JavaScript, MATLAB, Java, SQL, HTML/CSS
Frameworks/Libraries: React, Next.js, React Native, Expo, NumPy, Pandas, SciPy, Librosa, BeautifulSoup, Leaflet, FastAPI, CatBoost
Technologies: Cloud Run, Docker, Firebase, Firestore, Google Cloud Storage, Git, GitHub, Vercel, FFmpeg
Technical: Web Scraping, REST APIs, Data Processing, Sentiment Analysis, Full-Stack Development, Audio Synthesis, Digital Signal Processing

AWARDS

Midnight Hackathon: August 2026 - Mobile Track Winner
Developed *Punchd*, a loyalty reward system built on mobile to protect users' geographic and timestamp data using *Midnight*'s zero-knowledge privacy blockchains.